

TaCFPext-MO: Overflow-Triggered Multi-Superframe Order Adaptation for STREAM-E (Objective 4)

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1 Problem Statement

STREAM-E's Algorithm 1 ranks and allocates s total GTS slots among n competing child devices each multi-superframe cycle. When aggregate demand from Critical/Delay-driven (Ct/Dt) devices exceeds s , even top-ranked devices can receive $\text{gts_alloc}[i] = 0$: priority ranking has done all it can, and the packet is dropped. TaCFPext-MO prevents this by growing s itself, via `macMultisuperframeOrder` (MO), whenever the drop is sustained – and shrinking it back once the extra capacity is no longer earning its keep.

Standard constraints (IEEE 802.15.4-2024, §10.4.3)

$$0 \leq SO \leq MO \leq BO \leq 14 \quad (\text{non-extended DSME}) \quad (1)$$

$$SO \leq MO \leq BO + \text{len}(\text{BSN}) \quad (\text{macExtendedDsmeEnabled} = \text{TRUE}) \quad (2)$$

The superframe count per multi-superframe is $N = 2^{MO-SO}$. With `SlotsPerSF` = 7 CFP slots per superframe (matching STREAM-E Algorithm 3's $i = 0..6$ loop), total allocatable capacity is

$$N_{\text{GTS}}(MO) = \text{SlotsPerSF} \times 2^{MO-SO}. \quad (3)$$

Raising MO by 1 doubles capacity while leaving SO , BO , CAP timing, and the energy-aware extension's $E_{\text{gts}} = \text{gts_alloc} \times a\text{BaseSlotDuration} \times 2^{SO} \times E_{\text{rx}}$ completely unaffected.

2 Failure-Mode Analysis and Resolutions

Failure mode	Resolution
Oscillating drops evade a consecutive-run counter	Sliding-window sums growSum, shrinkSum over the last N_g/N_s cycles
Zero recent drops $\neq$ safe to shrink	Shrink gated on <i>both</i> low drop-frequency <i>and</i> low average utilization
Shrink right after a grow re-triggers instantly	Cross-cooldown: C_s applies to time-since-grow <i>and</i> time-since-shrink
Single-step growth too slow for a severe burst	Deficit-sized jump, capped by MAX_STEP
Shrink can silently drop the exact traffic this system protects	Ct/Dt never evicted; Rt/Nt reclaimed via standard §10.4.6.3 deallocation, not an internal wipe
Fixed MO_{max} constant can exceed the legal ceiling	$MO_{max} \leftarrow BO$ (or $BO + \text{len}(\text{BSN})$), derived at init
Detection cadence mismatched with Algorithm 1's cadence	Detection runs once per multi-superframe cycle
deficit mixed slot-counts with event-counts	Redefined as worst-case count of denied CD devices in-window (conservative lower bound)
Repeated $s \times 2 / s \div 2$ drifts under repeated resizes	Recompute $s = N_{\text{GTS}}(MO)$ from Eq. 3 on every commit
New CFP slots have no channel assigned yet	Growth commit re-runs Algorithm 3 over the appended range
Overlapping resize decisions can queue mid-flight	resizePending guard freezes decisions (history logging continues)
Bigger MO enlarges the SAB in every EB	Flagged as an empirical frame-size ceiling to measure in Cooja

3 Notation and Constants

- $CD = \{i : \text{class}[i] \in \{Ct, Dt\}\}$
- N_g, N_s – grow/shrink window lengths, in multi-superframe cycles ($N_s > N_g$)
- X_{high}, X_{low} – drop-cycle thresholds to trigger growth / permit shrink
- U_{low} – average utilization ceiling below which shrink is considered
- C_g, C_s – cooldowns in cycles ($C_s > C_g$)
- MAX_STEP – max ΔMO per resize event
- MO_{min} – initial deployment MO , hard-floored at SO
- $MO_{max} = BO$ (or $BO + \text{len}(\text{BSN})$ under extended DSME)

4 Consolidated Algorithm

Algorithm 5 TaCFPext-MO — Detection (runs once per multi-superframe cycle k , after Algorithm 1)

Require: $\text{gts_req}[\cdot], \text{gts_alloc}[\cdot]$ for cycle k ; $s = N_{\text{GTS}}(MO)$; ring buffers $\text{dropHist}, \text{utilHist}$ (size N_s); resizePending

Ensure: $\text{growSignal}, \text{shrinkSignal}, \text{deficit}$

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1: if  $\text{resizePending}$  then
2:   return (FALSE, FALSE,  $-$ ) ▷ freeze decisions while a change is in flight
3: end if
4:  $\text{drop}_k \leftarrow |\{i \in CD : \text{gts\_req}[i] = 1 \wedge \text{gts\_alloc}[i] = 0\}|$ 
5:  $\text{used}_k \leftarrow \sum_i \text{gts\_alloc}[i]$  ▷ all traffic classes
6:  $\text{util}_k \leftarrow \text{used}_k / s$ 
7:  $\text{push } \text{drop}_k \rightarrow \text{dropHist}; \text{ push } \text{util}_k \rightarrow \text{utilHist}$ 
8:  $\text{growSum} \leftarrow \sum(\text{last } N_g \text{ entries of dropHist})$ 
9:  $\text{shrinkSum} \leftarrow \sum(\text{last } N_s \text{ entries of dropHist})$ 
10:  $\text{avgUtil} \leftarrow \text{mean}(\text{last } N_s \text{ entries of utilHist})$ 
11:  $\text{deficit} \leftarrow \max(\text{last } N_g \text{ entries of dropHist})$ 
12:  $\text{growSignal} \leftarrow (\text{growSum} \geq X_{\text{high}}) \wedge (MO < MO_{\text{max}}) \wedge (k - \text{lastGrow} \geq C_g)$ 
13:  $\text{shrinkSignal} \leftarrow (\text{shrinkSum} \leq X_{\text{low}}) \wedge (\text{avgUtil} \leq U_{\text{low}}) \wedge (MO > MO_{\text{min}})$ 
    $\wedge (k - \text{lastShrink} \geq C_s) \wedge (k - \text{lastGrow} \geq C_s)$ 
14: return ( $\text{growSignal}, \text{shrinkSignal}, \text{deficit}$ ) ▷ mutually exclusive; grow takes priority

```

Algorithm 6 TaCFPext-MO — Growth

Require: $\text{deficit}, \text{current } MO, MO_{\text{max}}, \text{MAX_STEP}$

Ensure: pendingMO

```

1:  $\text{requiredCapacity} \leftarrow N_{\text{GTS}}(MO) + \text{deficit}$ 
2:  $\text{candidate} \leftarrow MO$ 
3: while  $N_{\text{GTS}}(\text{candidate}) < \text{requiredCapacity} \wedge \text{candidate} < MO_{\text{max}}$  do
4:    $\text{candidate} \leftarrow \text{candidate} + 1$ 
5: end while
6:  $\text{pendingMO} \leftarrow \min(\text{candidate}, MO + \text{MAX\_STEP}, MO_{\text{max}})$ 
7: if  $\text{pendingMO} = MO$  then
8:    $\text{SATURATED} \leftarrow \text{TRUE}; \text{return } \emptyset$  ▷ at ceiling; ranking degrades gracefully, no further
    $\text{MAC-layer mitigation}$ 
9: end if
10:  $\text{resizePending} \leftarrow \text{TRUE}; \text{ lastGrow} \leftarrow k$ 
11:  $\text{SCHEDULEUPDATE}(\text{pendingMO}, \text{APPEND})$  ▷ append-only  $\Rightarrow$  no remap of existing GTS
12: return  $\text{pendingMO}$ 

```

Algorithm 7 TaCFPext-MO — Shrink

Require: current MO , MO_{min} **Ensure:** pendingMO or ABORT

```
1: pendingMO  $\leftarrow$  max( $MO - 1$ ,  $MO_{min}$ )
2: victimSFs  $\leftarrow$  superframe indices removed when shrinking  $MO \rightarrow$  pendingMO  $\triangleright$  the truncated tail
3: if  $\exists$  active Ct/Dt GTS inside victimSFs then
4:   return ABORT  $\triangleright$  never evict Ct/Dt; re-check next eligible cycle
5: end if
6: reclaimSet  $\leftarrow$  active Rt/Nt GTS inside victimSFs
7: for all  $g \in$  reclaimSet do
8:   initiate standard DSME GTS deallocation (§10.4.6.3) toward  $g$ 's owner
9: end for
10: resizePending  $\leftarrow$  TRUE; lastShrink  $\leftarrow k$ 
11: SCHEDULEUPDATE(pendingMO, TRUNCATE)
12: return pendingMO
```

Algorithm 8 TaCFPext-MO — Broadcast & Resync

Require: pendingMO, mode $\in$ {APPEND, TRUNCATE}**Ensure:** committed MO , updated DSME PAN Descriptor / SAB

```
1: if mode = APPEND then
2:   newSAB  $\leftarrow$  currentSAB extended with empty entries for new superframes
3: else
4:   newSAB  $\leftarrow$  currentSAB restricted to surviving superframes  $\triangleright$  reclaimed slots already deallocated (Alg. 7)
5: end if
6: nextEB.macMultisuperframeOrder  $\leftarrow$  pendingMO
7: nextEB.DSMEPANDescriptor  $\leftarrow$  REBUILD(newSAB, pendingMO)
   on EB transmission at slot 0 of the next superframe (§10.4.3, atomic sync point):
8:    $MO \leftarrow$  pendingMO; currentSAB  $\leftarrow$  newSAB;  $s \leftarrow N_{GTS}(MO)$   $\triangleright$  Eq. 3, not incremental
9: if mode = APPEND then
10:   run Algorithm 3 (channel assignment) over the newly appended CFP slots
11: end if
12: clear dropHist  $\triangleright$  utilHist is never cleared
13: resizePending  $\leftarrow$  FALSE
```

5 Complexity

Detection is $O(n)$ per cycle (one pass over child devices, dominated by Algorithm 1 itself). The growth search loop is bounded by $\log_2(MO_{max} - MO)$ steps but is additionally capped by MAX_STEP, so it is $O(1)$ in practice. Shrink's feasibility check over victimSFs is $O(\text{SlotsPerSF}) = O(1)$ since SlotsPerSF = 7 is fixed by the standard. TaCFPext-MO therefore adds no asymptotic overhead to STREAM-E's existing $O(n)$ per-cycle cost.

6 Open Parameters for Empirical Tuning (Cooja)

Parameter	Suggested start	Notes
N_g	3–5 cycles	short enough to react before sustained loss
N_s	$3 \times N_g$	long enough to avoid premature shrink
X_{high}	$\lceil N_g/2 \rceil$	majority-of-window drop rate
X_{low}	0 or 1	near-zero tolerance before considering shrink
U_{low}	0.5–0.6	below this, the extra capacity is idle overhead
C_g	1–2 cycles	fast reaction
C_s	$\geq 2 \times C_g$	deliberately slower/cautious
MAX_STEP	2	bounds worst-case jump from one detection event